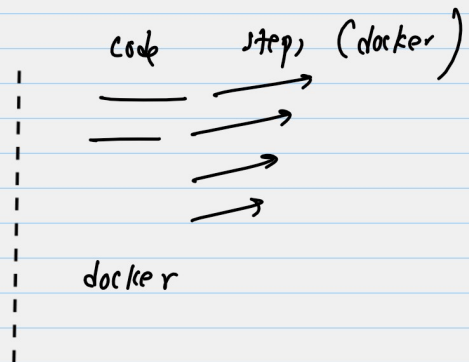


19-07-2026

Agenda:

Gradient Boosting & Xgboost classifier

fastapi → ml deployed
(another doc)



Regression → $\frac{1}{2}(y - \hat{y})^2$
↓
gradient
↓
residual

$\log \frac{p}{1-p} \leftarrow f_0$

Trees
↙ ↘ does not output prob
→ numbers / logit

logistic regression → linear reg + sigmoid

Dataset

ID	Size	Beds	price	label(y)
1	800	2	120	0
2	900	2	130	0
3	1000	3	150	0
4	1100	3	170	1
5	1200	4	200	1

price $\geq 170 \rightarrow \text{Yes} \mid 1 \text{ else No} \mid 0$

Positive labels (1)
total no of label

step 1: positive fraction $\hat{p} = \frac{2}{5} = 0.4$

step 2: log-odds initial prediction:

$$f_0 = \log \frac{p}{1-p} = \log \frac{0.4}{0.6}$$

$$\approx -0.4054$$

step 3: initial prob for every row

$$p_0 = \sigma(f_0) = 0.4$$

ID	Size	Beds	price	label(y)	P ₀
1	800	2	120	0	0.4
2	900	2	130	0	0.4
3	1000	3	150	0	0.4
4	1100	3	170	1	0.4
5	1200	4	200	1	0.4

logistic loss $L =$

$$y \log(p) + (1-y) \log(1-p)$$

where $p = \sigma(f)$

$$\text{gradient} = \frac{\partial L}{\partial f} = p - y$$

$$\begin{aligned} \text{hessian} &= p(1-p) \\ &= 0.4 \times 0.6 \\ &= 0.24 \end{aligned}$$

step 4: compute g & h for every row

ID	label(y)	P ₀	g = p - y	h
1	0	0.4	0.4	0.24
2	0	0.4	0.4	0.24
3	0	0.4	0.4	0.24
4	1	0.4	0.6	0.24
5	1	0.4	0.6	0.24

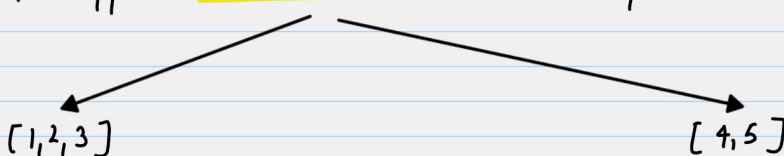
step 5: creation of stump

midpoint:

size: 850, 950, 1050, 1150
price: 125, 140, 160, 185
beds: 2.5, 3.5

for each possibilities

let's suppose size < 1050 is the best split



$$G_L = \sum g_i$$

$$G_R = \sum g_i$$

$$G_L = 0.4 + 0.4 + 0.4 = 1.2$$

$$G_R = -0.6 + -0.6 = -1.2$$

$$H_L = 0.24 \times 3 = 0.72$$

$$H_R = 0.24 \times 2 = 0.48$$

$$G_P = G_L + G_R$$

$$\begin{aligned} G_P &= 1.2 + (-1.2) = 0 \\ H_P &= 0.72 + 0.48 = 1.2 \end{aligned}$$

1) calculate g & h for each side (L & R)

2) compute parent $g_p = g_L + g_R$
 $h_p = h_L + h_R$

3) calculate gain:

$$\frac{1}{2} \left(\frac{g_L^2}{H_L + \lambda} + \frac{g_R^2}{H_R + \lambda} - \frac{g_P^2}{H_P + \lambda} \right)$$

where $\lambda = 0$ (pure GB)

4) calculate leaf output:

$$Y_{\text{side}} = -\frac{G_{\text{side}}}{H_{\text{side}}}$$

Gain : $\frac{1}{2} \left(\frac{G_L^2}{H_L} + \frac{G_R^2}{H_R} - \frac{G_P^2}{H_P} \right) \rightarrow \frac{1}{2} \left(\frac{(1.2)^2}{0.72} + \frac{(-1.2)^2}{0.48} - 0 \right)$

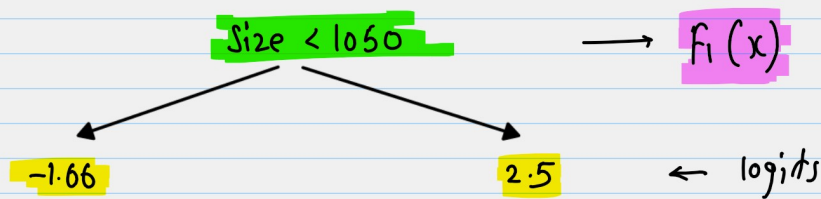
$\rightarrow \frac{5}{2} = 2.5 \rightarrow$ let's suppose this is the best gain

Step 6: calculate leaf output

G_L	H_L	G_R	H_R
1.2	0.72	-1.2	0.48

$y_L = \frac{-G_L}{H_L} = \frac{-1.2}{0.72} = -1.66$

$y_R = \frac{-G_R}{H_R} = \frac{-(-1.2)}{0.48} = 2.5$



Inference: input { size: 800, bed: 2, price: 120 }

$\downarrow$ left side

$f = f_0(x) + \eta \cdot f_1(x)$

$= -0.40546 + 0.1 \cdot (-1.66) \approx -0.5721$

$p_{\text{final/pred}} = \sigma(f) = \sigma(-0.5721) \approx 0.36$

Inference: input { size: 1800, bed: 2, price: 120 }

$f = f_0(x) + 0.1 \cdot f_1(x) \rightarrow$ right side

$= -0.40546 + 0.1 \cdot (2.5) \approx -0.15546$

$p_{\text{final/pred}} = \sigma(f) = \sigma(-0.15546) \approx 0.46$

ID	Size	Beds	price	label (y)	P_0	$g = p - y$	h	P_1
1	800	2	120	0	0.4	0.4	0.24	0.36
2	900	2	130	0	0.4	0.4	0.24	0.36
3	1000	3	150	0	0.4	0.4	0.24	0.36
4	1100	3	170	1	0.4	0.6	0.24	0.46
5	1200	4	200	1	0.4	0.6	0.24	0.46

next-model:

midpoint candidate columns: Size, Beds, Price

$$g: P_1 - y$$

$$h: P_1(1 - P_1)$$

follow same process

xgboost (classification)

ID	Size	Beds	price	label (y)	P_0	$g = p - y$	h
1	800	2	120	0	0.4	0.4	0.24
2	900	2	130	0	0.4	0.4	0.24
3	1000	3	150	0	0.4	0.4	0.24
4	1100	3	170	1	0.4	0.6	0.24
5	1200	4	200	1	0.4	0.6	0.24

Step (1) Initial model with log of odd (F_0)

$$\downarrow$$

$$P_0 = \sigma(F_0) = 0.4$$

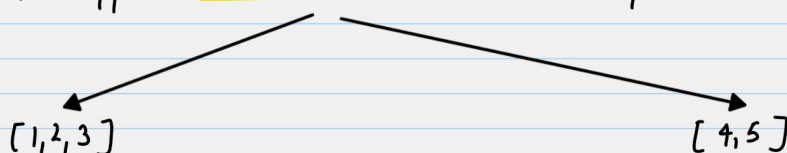
Step (2)

Use same logistic loss



Evaluate each candidate split

Let's suppose $\text{size} < 1050$ is the best split



$$G_L = \sum g_i \quad G_R = \sum g_i$$

$$G_L = 0.4 + 0.4 + 0.4 = 1.2$$

$$H_L = 0.24 \times 3 \rightarrow 0.72$$

$$G_R = -0.6 + -0.6 = -1.2$$

$$H_R = 0.24 \times 2 = 0.48$$

$$G_P = G_L + G_R$$

$$G_P = 1.2 + (-1.2) = 0$$

$$H_P = 0.72 + 0.48 = 1.2$$

$$Gain_{XGB} = \frac{1}{2} \left(\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{G_P^2}{H_P + \lambda} \right) - \gamma$$

$$\lambda = 1$$

$$\gamma = 0$$

Xgboost gain(λ, γ)

$$\lambda = 1, \gamma = 0$$

$$Gain = \frac{1}{2} \left(\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{G_P^2}{H_P + \lambda} \right) - \gamma$$

$$= \frac{1}{2} \left(\frac{1.44}{0.72 + 1} + \frac{1.44}{0.48 + 1} - 0 \right) - 0$$

$$\approx 0.90509$$

Gain (Gradient Boosting)

$$2.5$$

Gain (Xgboost)

$$0.9051$$

λ reduces the raw gain bcoz denominators are large

Xgboost leaf weights

$$w_{side} = - \frac{G_{side}}{H_{side} + \lambda}$$

$$w_L = \frac{-1.2}{0.72 + 1} = -0.697674$$

$$w_R = \frac{-(-1.2)}{0.48 + 1} = 0.81081$$

weights (Gradient Boosting)

L	R
-1.66	2.5

weights (Xgboost)

L	R
-0.6976	0.8108

input:

$$x = \text{size: 800} \rightarrow f_0(x) + \eta \cdot f_1(x)$$

$$p_1(\sigma(f(x)))$$

$$f(x) \rightarrow -0.40546 + 0.1 \cdot (-0.69767) = -0.4752$$

$$0.38$$

$$x = \text{size: 1800}$$

$$\rightarrow -0.40546 + 0.1 \cdot (0.8108) = -0.3243$$

$$0.41$$

ID	Size	Beds	price	label (y)	P_0	$g = p - y$	h	$P_1(\text{GB})$	$P_1(\text{XGB})$
1	800	2	120	0	0.4	0.4	0.24	0.36	0.38
2	900	2	130	0	0.4	0.4	0.24	0.36	0.38
3	1000	3	150	0	0.4	0.4	0.24	0.36	0.38
4	1100	3	170	1	0.4	0.6	0.24	0.46	0.41
5	1200	4	200	1	0.4	0.6	0.24	0.46	0.41

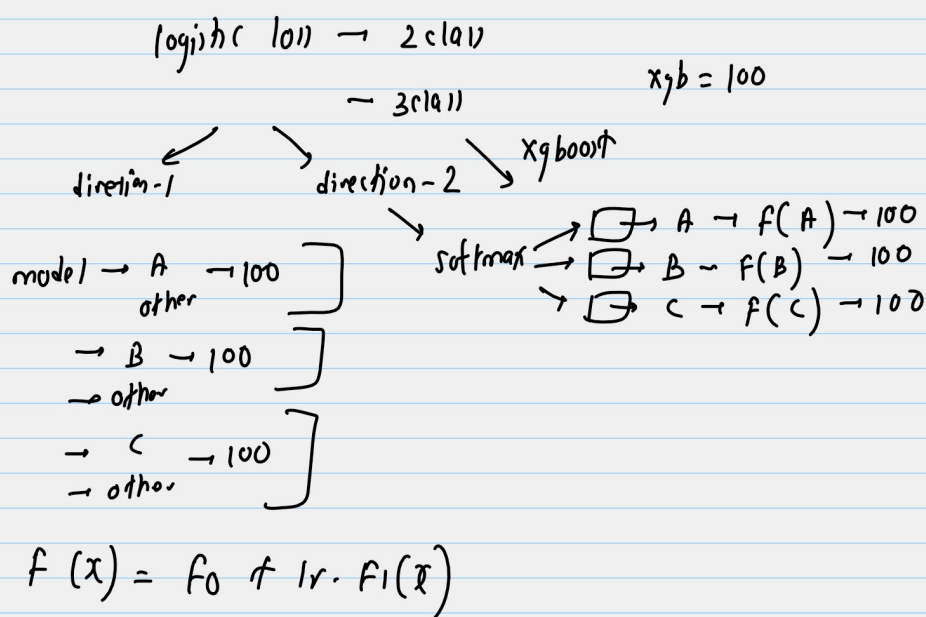
Method	leaf-weight (L)	leaf-weight (R)	$f_1(L)$	$f_1(R)$	$P_1(L)$	$P_1(R)$
GB ($\lambda=0$)	-1.666	2.5	-0.572	-0.155	0.36	0.46
XGB ($\lambda=1$)	-0.697	0.8108	-0.475	-0.324	0.38	0.41
					↓	↓
					0	1

GB (no λ) makes larger jumps (stronger corrections).

XGBoost with $\lambda=1$ makes smaller, more conservative updates (regularized), which often generalizes better.

learn $\rightarrow$ faster $\rightarrow$ train fast $\rightarrow$ less tree

learn $\rightarrow$ slow $\rightarrow$ train slow $\rightarrow$ more tree $\rightarrow$ precise



Machine Learning Completed